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Carbohydrate metabolism in women who used oral contraceptives containing levonorgestrel or desogestrel: a 6-month prospective study*

Alfred S. Luyckx, M.D.†‡
Ulysse J. Gaspard, M.D.§
Michèle A. Romus, M.D.§
Florin Grigorescu, M.D.||
Pierre De Meyts, M.D.||
Pierre J. Lefèbvre, M.D.¶

University of Liège, Liège and International Institute of Cellular and Molecular Pathology, Catholic University of Louvain, Brussels, Belgium

Blood glucose and pyruvate, plasma insulin, and glucagon levels as well as erythrocyte insulin receptors were measured during an oral glucose tolerance test in 38 normal women before and after 6 months' use of one of three new oral contraceptives containing low doses of 19 nortestosterone-derived progestogens, levonorgestrel, and desogestrel. A slight deterioration of glucose tolerance was observed, with the area under the glucose curve increasing by only 7%, 9%, and 12% after Ovidol (Aaciphar SA, Brussels, Belgium), Marvelon (Organon, SA, Brussels, Belgium), and Trigynon (Schering SA, Brussels, Belgium) administration, respectively. We did not find any argument in favor of the development of a state of insulin resistance in women using these compounds, because erythrocyte receptor binding was not modified and plasma insulin responses to glucose were decreased. The glucose-induced suppression of plasma glucagon levels seemed less effective for treatment with the desogestrel-containing preparations than with the levonorgestrel-containing oral contraceptives. Fertil Steril 45:635, 1986

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*Supported in part by Schering SA, Brussels, Belgium.
†Division of Clinical Pharmacology, State University of Liège.
‡Deceased on November 2, 1985.
§Department of Obstetrics and Gynecology, University of Liège.
||International Institute of Cellular and Molecular Pathology, Hormone and Metabolic Research Unit, Catholic University of Louvain.
¶Reprint requests: Professor Pierre J. Lefèbvre, Division of Diabetes, Institute of Medicine, Hôpital de Bavière, Boulevard de la Constitution, 66, B-4020 Liège, Belgium.

It is now generally believed that the estrogen ethinylestradiol (EE_2) has little effect on glucose or insulin circulating levels[1] and that the alterations of carbohydrate metabolism, which sometimes occur in women using combined estrogen-progestin oral contraceptives (OCs) are related primarily to the progestogen content of the pill.[2, 3] When studying three widely used 19 nortestosterone-derived progestogens, namely norethindrone, ethynodiol diacetate, and norgestrel or its active enantiomer levonorgestrel, it is obvious that all three steroids alter glucose tolerance

and that both blood glucose and plasma insulin levels in users are elevated.[4-6] These effects seem in part dose-related and potency-related[2]; accordingly, levonorgestrel, a potent compound, seems to be more effective in impairing glucose tolerance, either alone or in combination with EE_2,[2, 6] especially when administered in high dosages. Briggs and Briggs,[7] who compared different standard and low-dose EE_2-levonorgestrel combined OCs, clearly showed that the lowest dose preparations that contained these steroids, such as the triphasic combination, had minimal effects on carbohydrate metabolism. These data compare closely with the results of Spellacy et al.[3] obtained during treatment with a low-dose combination of EE_2 and norethindrone.

This study was designed to compare blood glucose (BG), pyruvate and plasma insulin, and glucagon levels during oral glucose tolerance tests performed before and after 6 months' use of three new oral contraceptives. Erythrocyte insulin receptor levels were studied also, because De Pirro et al.,[8] among others, seemed to demonstrate a significant decrease in insulin receptors in women using OCs. The OCs we studied all contained low doses of 19 nortestosterone-derived progestogens, levonorgestrel, and desogestrel, the latter being a new-deoxo 11β-methylene levonorgestrel derivative.

MATERIALS AND METHODS

SUBJECTS

Forty healthy young women whose clinical characteristics were comparable (Table 1) volunteered for the study. The protocol was approved by the Ethical Committee for Human Investigations of our university. None of the women had a family history of diabetes. All were within 15% of their ideal body weight and either never had used oral contraception previously or had stopped taking OCs at least 8 weeks before entering the study. After 3 days on a diet containing at least 250 gm of carbohydrates/day and an overnight fast, the volunteers underwent an oral glucose tolerance test (OGTT) at 8:00 A.M. This control test was performed 7 days before menstruation during a normal spontaneous cycle. Thereafter, each of the volunteers was randomly allocated to one of the three OCs, whose composition is given in Table 2: Trigynon (Schering SA, Brussels, Belgium), Ovidol (Aaciphar SA, Brussels, Belgium), or Marvelon (Organon SA, Brussels, Belgium). Because two patients who discontinued participation in the group received Ovidol, the final number of patients in each group was 13 (Trigynon), 11 (Ovidol), and 14 (Marvelon), respectively. On the sixth month of contraceptive use, each woman reported back to the laboratory for a second oral glucose tolerance test. The test was performed during one of the last 3 days of OC intake.

METHODS

Blood samples were drawn from an antecubital vein before and every 30 minutes (for 3 hours) after administration of 75 gm of glucose dissolved in 300 ml of water. On each sample, BG was determined with the use the hexokinase method adapted to the Technicon AutoAnalyzer (Compagnie Belge Technicon, Brussels, Belgium), blood pyruvate according to the method of Czok and Lamprecht,[9] plasma immunoreactive insulin according to the method of Hales and Randle,[10] plasma immunoreactive glucagon according to the method of Luyckx,[11] with the use of antiserum 30 K from R. H. Unger (Dallas, TX). For each group, the means, standard deviations, and standard errors of the means for all four parameters were calculated at each time of measurement. The areas under the curves also were calculated for each individual patient.

Erythrocyte insulin receptors were measured before and 6 months after OC use according to the method of Gambhir et al.[12] Briefly, a tracer ($\sim 1.7 \times 10^{-11}$ M) of labeled porcine insulin (high-performance liquid chromatography purified Tyr^{A14}-^{125}I-insulin) is incubated for 90 minutes at 15°C in the appropriate buffer[12] in a total volume of 500 μl containing a concentration of 4.4×10^9 erythrocytes/ml and increasing concentrations (0 to 10^4 ng/ml) of unlabeled porcine insulin. Bound and free insulin are separated by centrifugation.[12] Binding is corrected for nonspecific binding by subtracting bound values in the presence of 10^4 ng/ml of unlabeled insulin. No patient with hematologic abnormalities susceptible of changing the mean age of erythrocytes was included in the study; reticulocyte counts were normal in all patients. Although entire binding curves are studied, the results are expressed only in terms of the initial binding of tracer alone, which is a reflection of the product of affinity × receptor concentration.

Statistical analysis was done using the Student's *t*-test for paired data for comparison of the

Table 1. *Clinical Characteristics of the Patients*[a]

	Age	Height	% Ideal body weight		OC used
			Before	After	
	yrs	*cm*			
Group 1 (n = 11)	22.4 ± 0.5	167.3 ± 1.3	101.5 ± 3.2	99.3[b] ± 3.1	Ovidol
Group 2 (n = 14)	22.5 ± 0.4	167.0 ± 1.2	100.2 ± 2.1	99.7 ± 2.0	Marvelon
Group 3 (n = 13)	23.3 ± 0.8	166.7 ± 1.8	102.2 ± 3.2	99.3 ± 3.6	Trigynon

[a]Mean ± SEM.
[b]$P < 0.02$ (Student's t-test for paired data).

6-month data with the control tests in a given group and the Student's t-test for nonpaired data for comparison of the three groups either before or 6 months after treatment.

RESULTS

TOLERANCE AND WEIGHT CHANGES

Subjective tolerance was excellent in all subjects in the Marvelon and Trigynon groups. Breast tension and gastrointestinal disturbances occurred in the patients receiving Ovidol, which resulted in the interruption of the treatment in 2 of the 13 patients. Therefore, 38 of the 40 patients completed 6 months of treatment and were included in the statistical analysis. The percentage of ideal body weight decreased slightly in all three groups, this decrease being statistically significant only in the Ovidol group ($P < 0.02$) (Table 1).

BLOOD GLUCOSE

A comparison of the groups failed to reveal any statistically significant difference between the BG values at any time of the control test (Fig. 1); areas under the glucose tolerance curves (GA) were also similar (Table 3). After 6 months of OC use, mean BG values were slightly higher in each group, compared with the control test, but the difference was statistically significant only between the 90th and the 180th minute for the women receiving Trigynon (Fig. 2). In this group, GA increased significantly between the control test and the test performed under treatment (+ 12%, $P < 0.05$). The 7% and 9% increases in GA in the groups receiving Ovidol and Marvelon, respectively, failed to reach the level of statistical significance. Finally, group comparison at the end of the study period did not display significant changes in mean BG values before and during the OGTT or in the mean GA (Table 3).

PLASMA INSULIN

The mean plasma insulin values are illustrated in Figure 3. The mean areas under the insulin curves (IA) before and after treatment are given in Table 3. A modest decrease in mean plasma insulin was observed in all three groups after 6 months on OCs, but the difference was statistically significant only at 0, 90, and 150 minutes in the group receiving Marvelon. After treatment, a 22% to 25% *decrease* in the integrated insulin response (IA) was significant ($P < 0.05$ versus control test) for all the three groups, with no significant difference between the groups at any of the time intervals. Calculation of the ratio of insulin to glucose area (IA/GA) revealed a comparable decrease from control to posttreatment test (−31%, −29%, and −31% for Trigynon, Marvelon, and Ovidol, respectively).

BLOOD PYRUVATE

In the control test, blood pyruvate exhibited a modest increase during the course of the OGTT in all three groups. The amplitude of this rise was statistically similar after 6 months' use of Marvelon and Ovidol (Fig. 4). In the group receiving

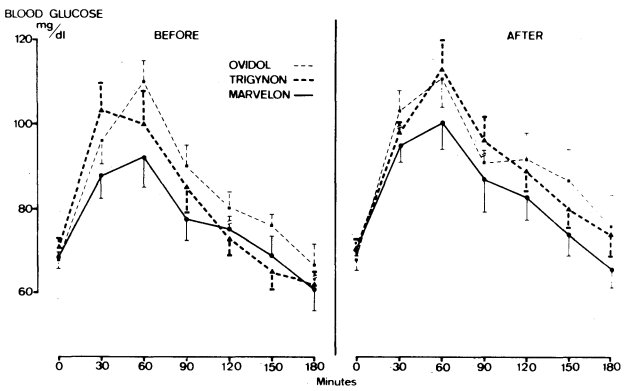

Figure 1
Group comparison of the BG levels measured during OGTT before and after 6 months on OC. Results are expressed as mean ± standard error of the mean (SEM).

Table 2. *OCs Used*

	Triphasic (Trigynon)	Sequential (Ovidol)	Monophasic (Marvelon)
Components	EE_2 0.03 mg + levonorgestrel (LNG) 0.05 mg (days 1–6)	EE_2 0.05 mg (days 1–7)	EE_2 0.03 mg + DOG 0.150 mg (days 1–21)
	EE_2 0.04 mg + LNG 0.075 mg (days 7–11)	EE_2 0.05 mg + desogestrel (DOG) 0.125 mg (days 8–21)	
	EE_2 0.03 mg + LNG 0.125 mg (days 12–21)		

Trigynon, blood pyruvate was higher at the 90th minute ($P < 0.005$) during treatment; the area under the pyruvate curve also was increased significantly (+ 26%, $P < 0.005$) when compared with the test carried out when the subjects were not using contraceptive agents. Nevertheless, the group comparisons of the areas under the pyruvate curve did not display any significant difference after treatment (Table 3).

PLASMA IMMUNOREACTIVE GLUCAGON

In the control test, the oral administration of glucose resulted in a parallel decrease in plasma immunoreactive glucagon in the three groups. The basal and postglucose plasma immunoreactive glucagon levels were not altered after 6 months' use of Trigynon (Fig. 5). In contrast, basal, and postglucose immunoreactive glucagon values were higher after Marvelon and Ovidol use; these changes were statistically significant for the second hour of the test. Because of the relatively large interindividual variations, the IRGAs were not significantly higher ($P < 0.1$) with OC use than during the control test (Table 3).

ERYTHROCYTE INSULIN RECEPTORS

Erythrocyte insulin receptor binding was in the normal range for premenopausal control women (which do not differ from age-matched control men) and were not significantly influenced by the use of any of the three contraceptives (Table 4).

DISCUSSION

Recent reports have emphasized the fact that the combined OCs that contain norgestrel, ethynodiol diacetate, or norethindrone may alter carbohydrate metabolism, especially in the case of preparations containing substantial amounts of the potent 19 nortestosterone steroid (levo)-norgestrel.[2] In contrast, preparations that contain low doses of EE_2 (0.030 to 0.040 mg)[3, 13] or levonorgestrel ($\leq$ 0.150 mg) restrict adverse effects on carbohydrate metabolism. This study aimed at investigating carbohydrate metabolism in women using three types of OCs, the first being a triphasic combination (Trigynon) containing a reduced dosage of levonorgestrel (1.925 mg/cycle) and the other two having a new progestogen, desogestrel, associated with either 0.03 mg of EE_2 in a combined monophasic pill (Marvelon) or 0.05 mg of EE_2 in a sequential type OC (Ovidol). A

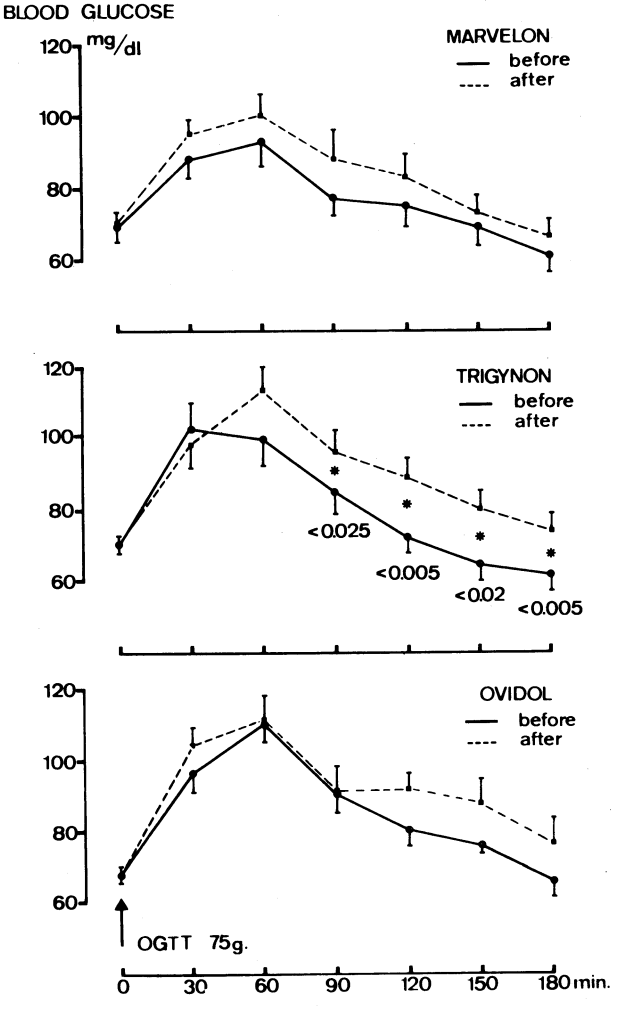

Figure 2
BG levels measured during OGTT before and after 6 months' use of Marvelon, Trigynon, or Ovidol. Results are expressed as mean ± SEM.

Table 3. *Comparison Between the Areas Under the Curves (During a 3-Hour OGTT) for Blood Glucose (GA), Plasma Insulin (IA), Blood Pyruvate (PA), and Plasma Immunoreactive Glucose (IRG A)*[a]

	Before	At 6 months	*t*	*P*
GA				
Ovidol	156.2 ± 5.0	167.5 ± 8.8	1.160	NS[b]
Trigynon	145.3 ± 5.8	162.5 ± 5.1	2.452	< 0.05
Marvelon	139.9 ± 7.3	152.6 ± 7.9	1.400	NS
IA				
Ovidol	114.2 ± 15.5	86.2 ± 7.7	2.499	< 0.05
Trigynon	84.4 ± 7.0	65.9 ± 2.8	2.456	< 0.05
Marvelon	91.6 ± 12.0	69.9 ± 8.0	2.293	< 0.05
PA				
Ovidol	95.8 ± 14.9	115.5 ± 6.7	1.394	NS
Trigynon	98.6 ± 9.1	124.4 ± 7.0	3.829	< 0.005
Marvelon	111.2 ± 14.7	113.1 ± 10.1	0.147	NS
IRG A				
Ovidol	98.0 ± 21.0	158.9 ± 42.0	2.004	NS
Trigynon	64.2 ± 15.2	82.8 ± 12.2	0.915	NS
Marvelon	103.8 ± 29.6	164.0 ± 47.5	2.099	NS

[a]Results are expressed as mean ± SEM in arbitrary units.
[b]NS, not significant.

slight deterioration of glucose tolerance was observed after 6 months' use of the three OCs, but this alteration reached the level of statistical significance only with Trigynon use. The GA increased by 12% with Trigynon, and by only 7% with Ovidol, and 9% with Marvelon. However, BG levels never reached pathologic levels in any of the women tested, and no statistical difference in glucose tolerance (mean values at all time intervals or GA) could be delineated between the three groups at the end of the 6-month study period. A similar degradation of the glucose tolerance curve (7% to 9% increase in GA) was reported after 3 to 15 months' use of a pill that contained 0.03 mg EE_2 and 0.150 mg levonorgestrel.[14] In contrast to our findings concerning Trigynon, Irsigler et al.[15] and Vermeulen and Thiery[16] could not delineate any significant change of the glucose tolerance curve after 3 to 6 months' use of the triphasic preparation. Clearly, the degradation of glucose tolerance was more marked after 18 months' use of 0.25 mg/day ethynodiol diacetate[5] or of 0.075 mg/day norgestrel alone.[6] Our observations with the two pills containing desogrestrel as progestins (Ovidol and Marvelon) are similar to those of Spellacy[3] and Spellacy et al.[13] The latter study showed no significant alterations in carbohydrate tolerance with a combination of 0.035 mg of EE_2 and 0.4 to 0.5 mg of norethindrone administered for 3 to 12 months.

It has been hypothesized that a decrease in the number and affinity of the insulin receptors is involved in the changes in carbohydrate metabolism reported after norgestrel administration,[8] whereas no alterations in insulin receptor capacity or affinity of monocytes were found in women using an OC containing a low dose of EE_2 associated with norethindrone.[17] However, we failed in this study to detect any change in number or affinity of erythrocyte insulin receptors, suggesting that neither the low doses of levonorgestrel used in the triphasic OC nor desogestrel induces a state of insulin resistance similar to that previously reported with other levonorgestrel containing pills or with progesterone itself.[14, 18] As reviewed by Spellacy,[2] the insulin response to glucose is increased in women who received 0.15 to 0.5 mg norgestrel associated with 0.03 mg EE_2 and unaltered under norethindrone (0.4 to 0.5 mg) plus EE_2 (0.035 mg). Surprisingly, we found a consistent 22% to 25% *reduction* in insulin response in the three groups investigated. These changes might reflect circannual variations in insulin response rather than drug-induced alterations in B-cell responsiveness and/or insulin sensitivity. Indeed, this 6-month study could have been influenced by the circannual changes in the plasma insulin response to glucose known to occur, at least in men.[19] Our results differ from those of Briggs and Briggs,[20] who reported, after 6 months' use of OCs, a significant increase in the insulin response with the pills containing desogestrel and no significant change after the use of Trigynon. In their study,[20] the GAs were unaltered under treatment with Marvelon and Trigynon and significantly increased after Ovidol (+ 18%). The reasons for the discrepancies between

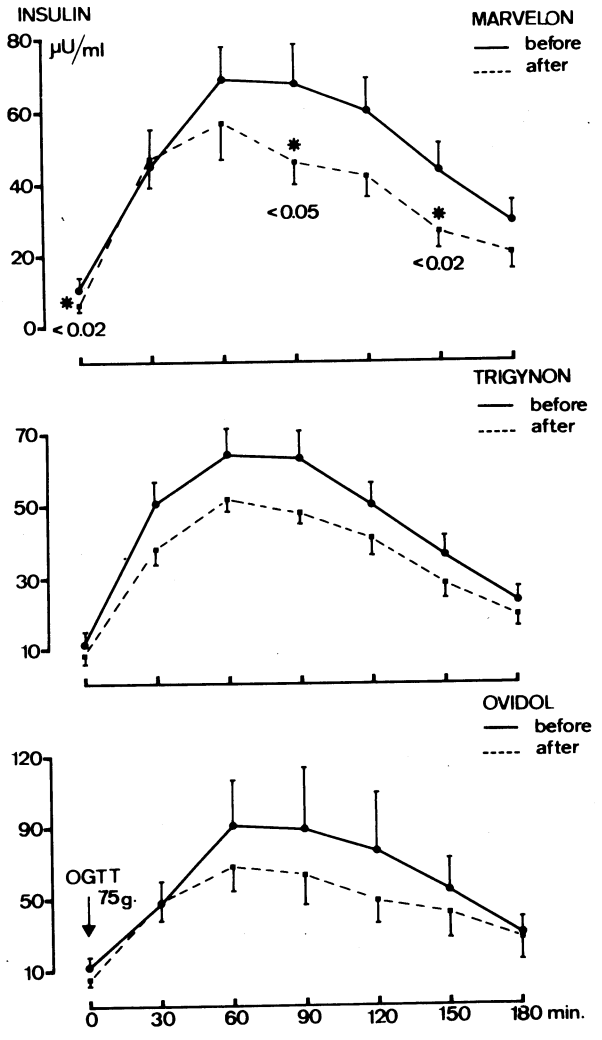

Figure 3
Plasma insulin levels measured during OGTT before and after 6 months' use of Marvelon, Trigynon, or Ovidol. Results are expressed as mean ± SEM.

these results and ours remains unexplained. Another point of interest in our study is that the rise in blood pyruvate usually observed during OGTT was unaltered by the contraceptives containing desogestrel but increased by 26% ($P < 0.005$) after use of Trigynon, which contains levonorgestrel. Whether this change is directly relevant to the mechanism(s) by which OCs alter glucose tolerance remains to be established, but such increases in blood pyruvate were also found after administration of corticosteroids and estrogen-progestogen therapy.[21]

Very few studies have been devoted to the influence of OCs on plasma glucagon. This study demonstrates that the triphasic administration of EE_2 and levonorgestrel does not modify either the fasting glucagon levels or the decline normally occurring after administration of glucose. These results are in agreement with the recent report of Gossain et al.,[22] which show that plasma glucagon concentrations in response to oral and intravenous glucose were not modified in healthy young women who used a combination of mestranol and norethindrone. In contrast, our results demonstrate that, in the two groups of patients using desogestrel as progestogen (Ovidol and Marvelon), plasma glucagon levels were slightly (but not significantly) higher after an overnight fast and that the maximum decline in the concentration of glucagon during OGTT was signifi-

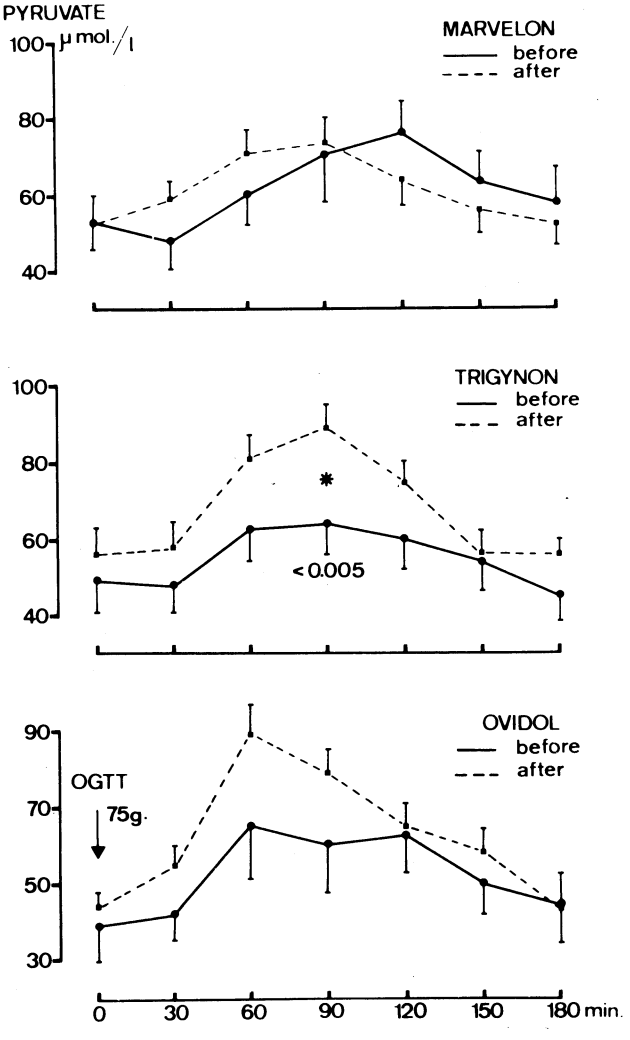

Figure 4
Blood pyruvate levels measured during OGTT before and after 6 months' use of Marvelon, Trigynon, or Ovidol. Results are expressed as mean ± SEM.

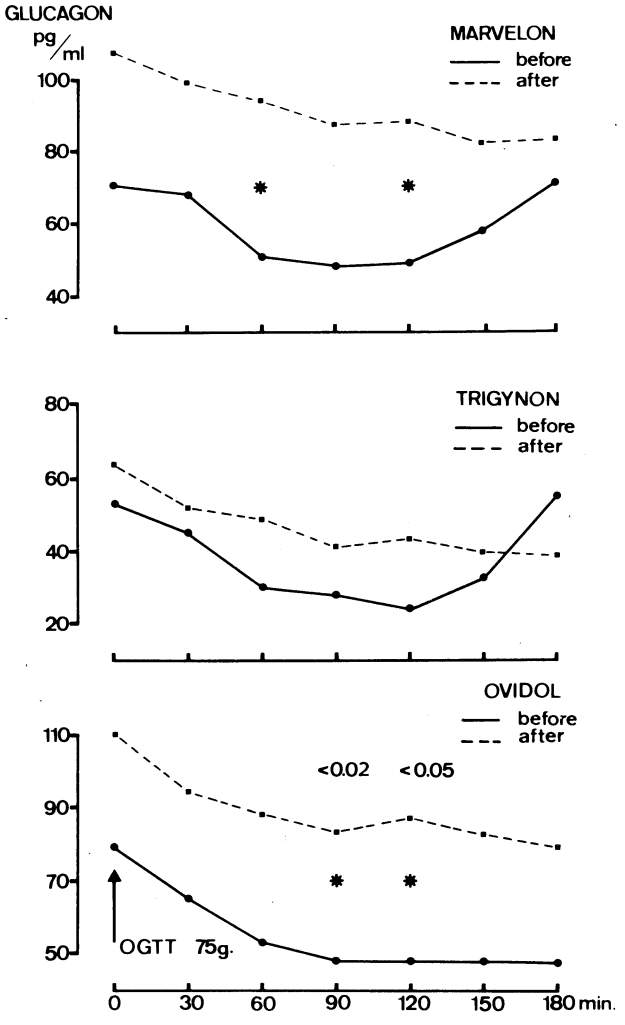

Figure 5
Plasma glucagon levels measured during OGTT before and after 6 months' use of Marvelon, Trigynon, or Ovidol. Results are expressed as mean; due to large interindividual variations, values of SEM were omitted.

cantly smaller when the subjects were using OCs than when they were not using them. Mandour et al.[23] showed that estrogens caused a lowering of the circulating glucagon levels but progesterone resulted in an increase in plasma glucagon concentrations. The balance between the two effects probably explains the discrepancies found in the various studies performed in the users of OCs, as these increase,[4] decrease,[25] or induce no appreciable changes in glucagon circulating levels.[22]

In conclusion, over a period of 6 months, the three new OC agents used in this study produced only small (Trigynon) or no (Ovidol, Marvelon) deterioration of the glucose tolerance curve in healthy women. However, it should be stressed

Table 4. *Erythrocyte Receptor Binding*[a]

OCs used	*n*	Pretreatment cycle[b]	Sixth cycle of OC use
Triphasic (EE_2 + LNG)	13	8.39 ± 1.19	8.36 ± 1.38
Monophasic (EE_2 + DOG)	14	7.99 ± 1.33	8.60 ± 1.98
Sequential (EE_2 + DOG	11	7.85 ± 2.02	7.74 ± 1.05

[a]Bound ^{125}I-insulin in percent of total.
[b]Mean ± standard deviation; changes are not statistically significant.

that no pathologic value of BG was reached at any time during OGTTs performed during use of oral contraception. Moreover, group comparison at the end of the study period did not delineate significant differences, either in mean BG values during OGTT or in the mean areas under the BG curves when the OCs under investigation were compared. We did not find any argument in favor of the development of a state of insulin-resistance in women using these compounds, because erythrocyte receptor function was not modified and plasma insulin responses to glucose were decreased. Finally, the OGTT-induced suppression of plasma glucagon levels seemed less effective during treatment with the desogestrel-containing preparations (Ovidol, Marvelon) than with the levonorgestrel-containing OC (Trigynon).

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